

**5-8K**

# SOLARATOR SERIES

Experience Uninterrupted Power with Solis Energy Storage Inverters

## S6-EH2P(5-8)K02-SV-YD-L

Split Phase | Low Voltage

- Supports PV input up to 160% of the inverter's rated DC power, maximising solar utilisation
- Supports up to 21A PV input current, compatible with future higher-power PV modules
- 200% overload for 10s in off-grid mode, ensuring stable startup of motors, water pumps and air conditioners
- Seamless on/off-grid switching in under 10ms, guaranteeing an uninterrupted power supply
- Support split-phase unbalanced output, each phase supports output max. 50% rated inverter power
- More battery options, compatible with any battery (between 40V-60V)
- Supports existing PV grid-tied power connection for export control and off-grid use
- Multiple methods of generator connection and auto control, enabling flexible local deployment
- Support max 6 units in parallel, for expanding system capacity
- Smart Load management, extending backup time for critical loads
- Customizable battery backup level for uninterrupted power
- PV only off grid mode, to reduce the initial investment cost
- 7-inch industrial-grade touchscreen, providing a larger, user-friendly interface for local operation
- Ingress protection IP66, adapting to harsh ambient conditions

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## DATASHEET

| Models                                          | 5K              | 6K                                                                                                            | 7.5K               | 8K              |
|-------------------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------|--------------------|-----------------|
| Input DC (PV side)                              |                 |                                                                                                               |                    |                 |
| Recommended max. PV array size                  | 10 kW           | 12 kW                                                                                                         | 16 kW              | 16 kW           |
| Max. usable PV input power                      | 8 kW            | 9.6 kW                                                                                                        | 12.8 kW            | 12.8 kW         |
| Max. input voltage                              |                 |                                                                                                               | 500 V              |                 |
| Rated voltage                                   |                 |                                                                                                               | 330 V              |                 |
| Start-up voltage                                |                 |                                                                                                               | 90 V               |                 |
| MPPT voltage range                              |                 |                                                                                                               | 90 - 435 V         |                 |
| Max. input current                              | 21 A / 21 A     | 21 A / 42 A                                                                                                   | 42 A / 42 A        | 42 A            |
| Max. short circuit current                      | 24 A / 24 A     | 24 A / 48 A                                                                                                   | 48 A / 48 A        | 48 A            |
| MPPT number / Max. input strings number         | 2 / 2           | 2 / 3                                                                                                         |                    | 2 / 4           |
| Battery                                         |                 |                                                                                                               |                    |                 |
| Battery type                                    |                 |                                                                                                               | Li-ion / Lead-acid |                 |
| Battery voltage range                           |                 |                                                                                                               | 40 - 60 V          |                 |
| Max. charge / discharge power                   | 5 kW            | 6 kW                                                                                                          | 7.5 kW             | 8 kW            |
| Max. charge / discharge current                 | 112 A           | 135 A                                                                                                         | 190 A              | 190 A           |
| Communication                                   |                 |                                                                                                               | RS485 / CAN        |                 |
| Output AC (Grid side)                           |                 |                                                                                                               |                    |                 |
| Rated output power                              | 5 kW            | 6 kW                                                                                                          | 7.5 kW             | 8 kW            |
| Rated grid voltage                              |                 | L+N+PE/2L+PE, 220 V; L1/L2/N(PE), 120(L-N)/240(L1-L2)VAC                                                      |                    |                 |
| Rated grid frequency                            |                 | 50 Hz / 60 Hz                                                                                                 |                    |                 |
| Rated grid output current                       | 20.8 A / 22.7 A | 25 A / 27.3 A                                                                                                 | 31.3 A / 34.1 A    | 33.3 A / 36.4 A |
| Max. output current                             | 22.9 A / 25 A   | 27.5 A/ 30 A                                                                                                  | 31.3 A / 34.1 A    | 36.7 A / 40 A   |
| Power factor                                    |                 | > 0.99 (0.8 leading - 0.8 lagging)                                                                            |                    |                 |
| THDi                                            |                 | < 3%                                                                                                          |                    |                 |
| Input AC (Grid side)                            |                 |                                                                                                               |                    |                 |
| Input voltage range                             |                 | L+N+PE/2L+PE, 220 V; L1/L2/N(PE), 120(L-N)/240(L1-L2)VAC                                                      |                    |                 |
| Max. input current                              | 35 A            | 40 A                                                                                                          | 50 A               | 50 A            |
| Frequency range                                 |                 | 45 - 55 Hz / 55 - 65 Hz                                                                                       |                    |                 |
| Input Generator                                 |                 |                                                                                                               |                    |                 |
| Max. input power                                | 5 kW            | 6 kW                                                                                                          | 7.5 kW             | 8 kW            |
| Max. input current                              | 20.8 A / 22.7 A | 25 A / 27.3 A                                                                                                 | 31.3 A / 34.1 A    | 33.3 A / 36.4 A |
| Rated input frequency                           |                 | 50 Hz / 60 Hz                                                                                                 |                    |                 |
| Output AC (Back-up)                             |                 |                                                                                                               |                    |                 |
| Rated output power                              | 5 kW            | 6 kW                                                                                                          | 7.5 kW             | 8 kW            |
| Max. apparent output power                      |                 | 2 times of rated power, 10 s                                                                                  |                    |                 |
| Back-up switch time                             |                 | < 10 ms                                                                                                       |                    |                 |
| Rated output voltage                            |                 | L1/L2/N(PE), 127 V / 220 V; L1/L2/N(PE), 120(L-N)/240(L1-L2)VAC                                               |                    |                 |
| Rated frequency                                 |                 | 50 Hz / 60 Hz                                                                                                 |                    |                 |
| Rated output current                            | 20.8 A / 22.7 A | 25 A / 27.3 A                                                                                                 | 31.3 A / 34.1 A    | 33.3 A / 36.4 A |
| Max. AC passthrough current                     | 35 A            | 40 A                                                                                                          | 50 A               | 50 A            |
| THDv (@linear load)                             |                 | < 3%                                                                                                          |                    |                 |
| Efficiency                                      |                 |                                                                                                               |                    |                 |
| Max. efficiency                                 |                 | 96.0%                                                                                                         |                    |                 |
| EU efficiency                                   |                 | 94.5%                                                                                                         |                    |                 |
| BAT charged by PV / AC max. efficiency          |                 | 95.0%                                                                                                         |                    |                 |
| BAT discharged to AC max. efficiency            |                 | 93.7%                                                                                                         |                    |                 |
| Protection                                      |                 |                                                                                                               |                    |                 |
| Surge protection                                |                 | Yes                                                                                                           |                    |                 |
| Ground fault monitoring                         |                 | Yes                                                                                                           |                    |                 |
| Integrated AFCI 2.0                             |                 | Yes                                                                                                           |                    |                 |
| DC reverse-polarity protection                  |                 | Yes                                                                                                           |                    |                 |
| Protection class / Over voltage category        |                 | I / II (PV and BAT), III (MAINS and BACKUP and GEN)                                                           |                    |                 |
| Dados gerais                                    |                 |                                                                                                               |                    |                 |
| Max. allowable phase imbalance (grid & back-up) |                 | 50%                                                                                                           |                    |                 |
| Dimensions (W × H × D)                          |                 | 390 × 620 × 237 mm                                                                                            |                    |                 |
| Weight                                          |                 | 28.8 kg                                                                                                       |                    |                 |
| Topology                                        |                 | Transformerless                                                                                               |                    |                 |
| Self-consumption (night)                        |                 | < 20 W                                                                                                        |                    |                 |
| Operating ambient temperature range             |                 | -25 ~ +60°C                                                                                                   |                    |                 |
| Relative humidity                               |                 | 0 - 100%                                                                                                      |                    |                 |
| Ingress protection                              |                 | IP66 (TYPE 4X)                                                                                                |                    |                 |
| Noise emission (typical)                        |                 | < 55 dB(A)                                                                                                    |                    |                 |
| Cooling concept                                 |                 | Intelligent fan-cooling                                                                                       |                    |                 |
| Max. operation altitude                         |                 | 4000 m                                                                                                        |                    |                 |
| Grid connection standard                        |                 | Inmetro140, IEEE 1547                                                                                         |                    |                 |
| Safety / EMC standard                           |                 | IEC/EN 61000-6-1/2/3/4, IEC/EN 62109-1, IEC/EN 62109-2, UL 1741, UL 1699B, FCC                                |                    |                 |
| Features                                        |                 |                                                                                                               |                    |                 |
| DC connection                                   |                 | MC4 plug (PV port) / Terminal Block (BAT port)                                                                |                    |                 |
| AC connection                                   |                 | OT terminal                                                                                                   |                    |                 |
| Display                                         |                 | 7.0" LCD display & Bluetooth + APP                                                                            |                    |                 |
| Communication interface                         |                 | Standard: Wi-Fi+LAN+Bluetooth, CAN-BMS, CAN-Parallel×2, RS485-Meter, RS485, DRM, DI, DO×2, Optional: 2G/3G/4G |                    |                 |